Answers and talk notes

The Distributive Property

What your student is learning

Tonight we use the distributive property on a real shopping problem — and show that the factored and expanded forms give the same total.

Answers

Ask your student to explain before you look.

1. **A.** 3(4 + 2). One kit costs 4 + 2, so 3 kits cost 3(4 + 2).
2. **C.** 45. 6 + 3 = 9 and 5 × 9 = 45. Also 5 × 6 + 5 × 3 = 30 + 15 = 45.
3. **A.** 7(4 + 5). Both terms have a factor of 7, so pull it out: 7(4 + 5).
4. **D.** 8a + 16. 8 × a = 8a and 8 × 2 = 16, so 8a + 16.
5. **B.** $42. One bag costs 4 + 3 = $7, and 6 × 7 = $42. Also 6 × 4 + 6 × 3 = 24 + 18 = $42.

Talk about it

Do both ways give the same total? Which way was easier for you, and why?

Home mission

A different situation: pretend you buy 5 snack packs. Each pack has a $3 box of crackers and a $2 juice. Say the cost two ways out loud. Try 5(3 + 2) and 5 × 3 + 5 × 2. Which way is easier in your head?

When they are stuck, ask

- Did the outside number reach both things inside?
- What does one kit cost? Now how many kits?
- Can you get the same total the other way?

Try not to say

- Multiply the outside number by the first price only.
- The two methods give different answers.
- You only need one method, skip the check.